Zambia: Debt Sustainability Assessment

Introduction

1. **This appendix reports the results of a debt sustainability assessment for Zambia conducted jointly by Fund and World Bank staff using the LIC DSA framework.** The baseline scenario extends the medium-term macroeconomic framework of the PRGF-supported program to 2025. In addition to standard alternative scenarios and stress tests two scenarios were considered: one that involves additional borrowing from power sector investment over 2008-11 and another that incorporates potential debt relief under the MDRI. In addition to external debt, the assessment also takes into account the implications of the outlook for domestic debt for fiscal sustainability.

2. **Zambia's external debt sustainability outlook has greatly improved with the attainment of the HIPC Initiative completion point in April 2005.** With the full delivery of HIPC Initiative assistance, as well as additional bilateral debt relief, Zambia's external debt is estimated at $3.8 billion in nominal terms at end-2004, corresponding to $1.5 billion in NPV terms, or 75 percent of exports. The bulk of this debt—almost 90 percent—is owed to multilateral institutions, in particular the Fund, IDA and the African Development Bank. Domestic debt, at 22 percent of GDP in NPV terms at end-2004, represents an additional element of risk to debt sustainability in Zambia, but this risk is mitigated by the government's commitment to limiting domestic financing to 0.5 percent of GDP per year over the long term. Fund and World Bank staff are of the view that Zambia is at low risk of debt distress going forward.

Baseline Scenario

3. **Policy-based indicative thresholds for debt burden indicators are a key feature of the LIC DSA framework**. The thresholds are based on the empirical finding that low-income countries with stronger policies and institutions tend to have higher debt carrying capacity. Zambia is classified as a medium performer in regard to policies and institutions, for which the relevant thresholds for debt burden indicators are 40 percent for the NPV of debt-to-GDP ratio, 150 percent for the NPV of debt-to-exports ratio, and 20 percent for the debt service-to-exports ratio.

4. **Zambia's external debt burden indicators are well below the policy-based thresholds of the LIC DSA framework in the baseline scenario and gradually improve over time.** The baseline scenario is based on an extension of the macroeconomic framework of the PRGF-supported program (Box A1). It assumes a continuation of Zambia's relatively strong recent performance, but incorporates a projected deterioration in the terms of trade as copper prices ease from their recent historic highs and a tapering off of output growth in the mining sector. In this scenario, the NPV of debt-to-GDP ratio is projected to decline gradually from 23 percent at end-2005 to 16 percent by 2010 and 10 percent by 2025 (Table A1 and Figure A1). The NPV of debt-to-exports ratio is also projected to decline, although somewhat more slowly, from 66 percent at end-2004 to 59 percent by 2010 and 45 percent by 2025, reflecting the slower growth of exports than of GDP. The debt service-to-exports ratio is

projected to fluctuate between 3-5 percent until 2010 and thereafter decline to 1 percent. In the baseline, the debt burden indicators remain significantly below their CPIA-linked thresholds for a medium performer on the quality of policies and institutions such as Zambia—40 percent for the NPV of debt-to GDP ratio, 150 percent for the NPV of debt-to-exports ratio, and 20 percent for the debt service-to-exports ratio—throughout the projection period.

Box 1. Macroeconomic Assumptions for Baseline Scenario, 2005-06

The baseline scenario is an extension of the macroeconomic framework that underlies the PRGF-supported program. It assumes that the government will continue to pursue sound macroeconomic and structural policies and that economic performance will be favorable.

- Real GDP grows by 6 percent per year on average.
- Inflation declines to 5 percent by 2007 and remains stable thereafter.
- The real effective exchange rate appreciates gradually—by about 2 percent per year.
- Exports growth is projected to be strong over the medium term, but to then moderate for a few years as metals output growth slows. Thereafter, export growth averages about 8 percent a year in dollar terms, supported by a rising share of nontraditional exports.
- Import growth is assumed to be strongly linked to GDP growth over the long run.
- Following the sharp recent improvement, the terms of trade are assumed to deteriorate gradually over the medium-term and then stabilize.
- The current account deficit (excluding grants) declines to 10 percent of GDP by 2010 and to about 8 percent by 2025.
- Government revenues remain stable over the medium term at about 18.5 percent of GDP, and thereafter rise gradually to 20 percent of GDP. External grants, both project and program, are stable at around 6 percent of GDP. Expenditures rise to 25 percent of GDP and remain stable thereafter. Domestic financing falls to ½ percent of GDP by 2008 and remains at that level.
- Foreign direct investment declines somewhat from its recent level of around 5 percent of GDP, associated with strong investment in the mining sector, to about 2 percent over the long run.
- External borrowing rises in nominal terms but declines gradually, relative to GDP, from its current level of about 2 percent to about 1 percent over the long run. All external borrowing is assumed to be concessional.

Standard Alternative Scenarios and Stress Tests

5. **Zambia's favorable external debt sustainability outlook is fairly robust to a deterioration in economic performance and exogenous shocks**. In a scenario that sets key variables equal to their averages over 1994-2004, and therefore takes into account the relatively poor performance of the Zambian economy during the earlier part of that period, the NPV of debt-to-GDP ratio declines throughout the projection period, albeit at a slower pace than in the baseline (Table A2). The NPV of debt-to-exports ratio, however, rises initially to 74

percent by 2010 and the declines gradually to 65 percent by 2025. The debt service-to-exports ratio is marginally higher than in the baseline.

6. **All but one of the standardized stress tests exhibit relatively benign debt dynamics**: all debt burden indicators remain well below relevant thresholds throughout. The exception is a shock to exports in 2006-07—modeled as export value growth at its historical average less one standard deviation, to allow for the possibility that copper prices fall sharply from their current elevated levels. In this scenario, the NPV of debt-to-exports ratio rises sharply in 2007, to above 150 percent where it stays through 2010. However, by 2015 the NPV of debt-to-exports ratio is projected to fall to 137 percent and to 94 percent by 2025. The NPV of debt-to-GDP and the debt service-to exports ratio also rise temporarily to peaks of 28 percent and 10 percent, respectively, well below the relevant thresholds.

Power Sector Investment Scenario

7. **Substantial investment in power generation financed by external borrowing on commercial terms has only a limited impact on Zambia's debt burden indicators**. As discussed in the staff report, ZESCO aims to expand its power generating capacity significantly over the medium term, with investment needs possibly financed on commercial terms. An alternative scenario that assumes external borrowing on nonconcessional terms—8 percent interest rate, 5 year grace, 17 year maturity—of 1 billion over 2008-2011 was prepared. The impact of this borrowing on debt and debt service ratios relative to the baseline is similar in magnitude to that of the historical scenario. Debt burden indicators never come close to their respective threshold.

MDRI Scenario

8. **Implementation of the MDRI would further improve Zambia's debt sustainability outlook**. The MDRI would cancel Zambia's obligations to the Fund, IDA, and the AfDF as of end-2004. This would lower Zambia's projected end-2006 nominal external debt from $3.6 billion to $0.5 billion. The reduction in the NPV of external debt-to-GDP ratio would be equally marked with the end-2006 value declining from 20 percent to 4 percent, while the NPV of external debt to export ratio would fall from 63 percent to 14 percent. These ratios would rise gradually over the long run as new borrowing comes to dominate the debt dynamics. The debt ratios remain low throughout relative to the relevant debt burden thresholds. The debt service-to-export ratio declines quickly to 2 percent and thereafter settles around 1 percent.

Fiscal Sustainability

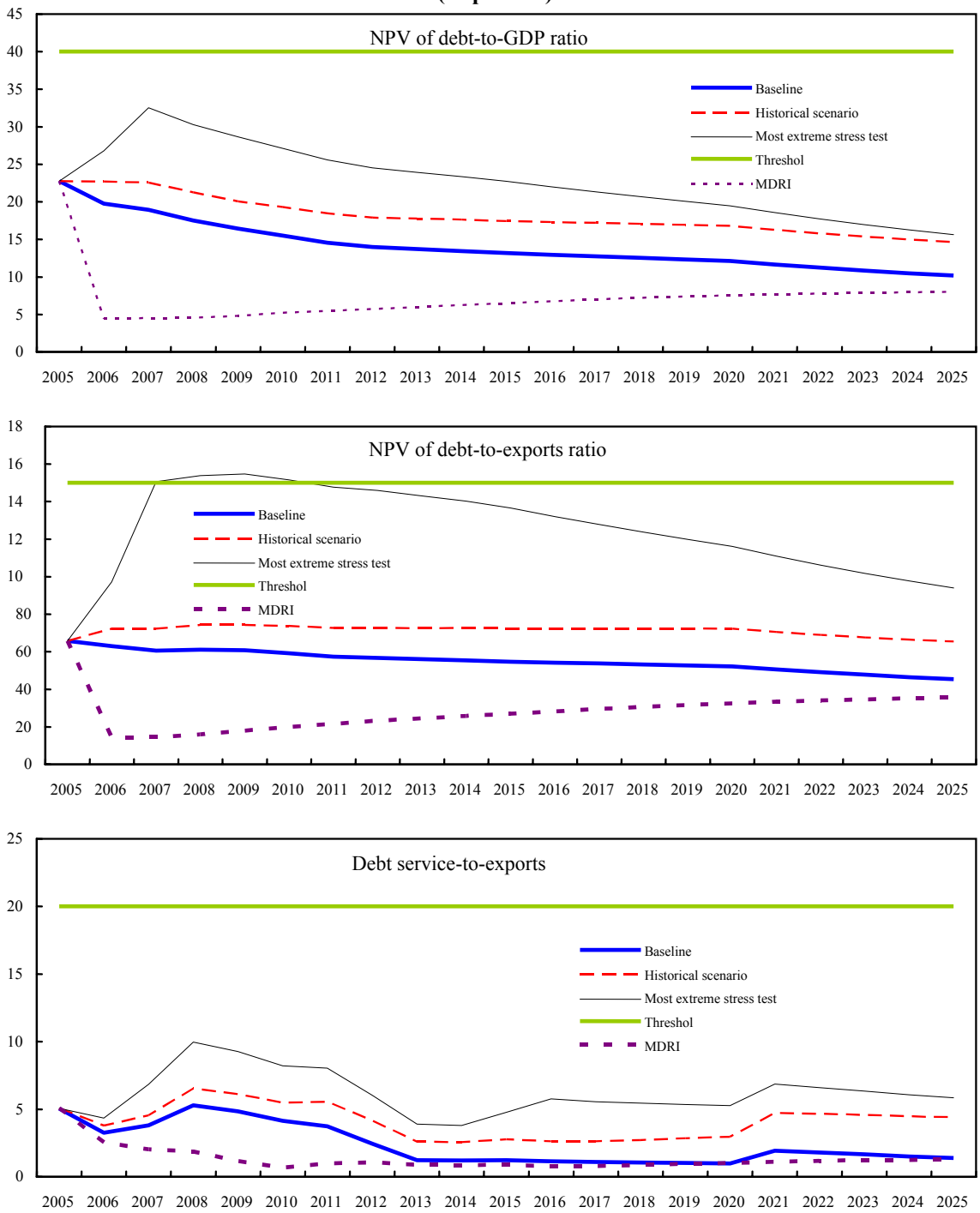
9. **Domestic debt accounts for a substantial share of overall public debt in Zambia**. Domestic debt is projected to be about 19 percent of GDP at end-2005, a sizable portion of which is in short-term treasury bills (Table A3 and Figure A2). Under the baseline scenario, which builds on the successful fiscal adjustment that the government has undertaken in recent years and incorporates its objective of limiting domestic financing to ½ percent of GDP per year over the long run, domestic debt is projected to decline to 12 percent of GDP by 2010 and 6 percent by 2025. The domestic debt-to-revenue ratio is projected to decline from 80 percent in 2005 to 50 percent by 2010 and 24 percent in 2025. Thus, in the baseline, the share of domestic debt in total public debt is projected to decline over time.

10. **Domestic debt is an additional element of risk to fiscal sustainability in Zambia.** Under standard alternative scenarios and stress tests, the projected debt dynamics are less favorable, sometimes significantly so (Table A4). The least favorable case is represented by a scenario in which key variables, including real GDP growth and the primary balance, are set at their historical averages, and the resulting financing requirement is met by additional domestic borrowing. In this scenario, the ratio of domestic debt to GDP is stable at 16-17 percent of GDP until 2010 before rising to 23 percent of GDP by 2015 and 26 percent by 2025. Relative to government revenue, the NPV of domestic debt is projected to ease from 70 percent in 2005 to 65 percent in 2010 before rising to 85 percent in 2015 and 123 percent in 2025. Moreover, the cost of servicing overall public debt in this scenario rises from a low of 13 percent of revenue in 2006-07 to 18 percent by 2010, 25 percent by 2015 and 39 percent by 2025.

Conclusion

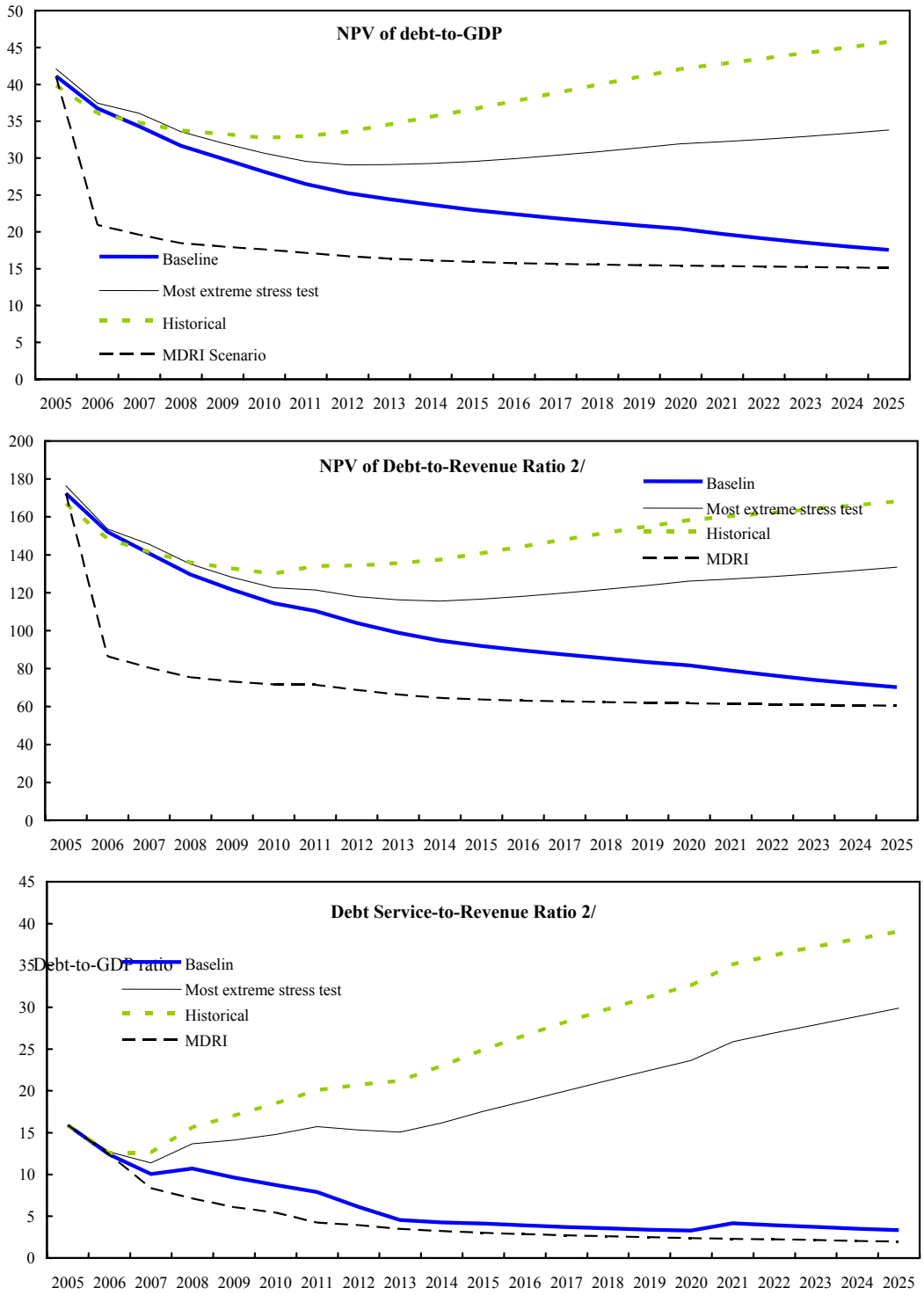
11. On the basis of the debt sustainability analysis that has been conducted, Fund and World Bank staff are of the view that Zambia is at a low risk of debt distress. Application of the LIC DSA framework to Zambia indicates that Zambia's external debt sustainability outlook is favorable under most scenarios, although it is vulnerable to a sharp setback to export earnings. Domestic debt and unfunded government liabilities represent some risks, although these stem to some extent more from the short maturity of the external debt than its volume. Implementation of the MDRI would further greatly strengthen Zambia's debt sustainability outlook and reduce Zambia's vulnerability to shocks.

Figure A1. Zambia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2005-2025 (In percent)



Source: Staff projections and simulations.

Figure A2.Zambia: Indicators of Public Debt Under Alternative Scenarios, 2005-2025 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.

Table A1. Zambia: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual	Historical Average 6/	Standard Deviation 6/	Estimate					Projections				
	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
External debt (nominal) 1/	**70.3**			**51.0**	**43.1**	**41.2**	**38.6**	**36.5**	**34.0**		**25.5**	**19.2**	
o/w public and publicly guaranteed (PPG)	70.3			51.0	43.1	41.2	38.6	36.5	34.0		25.5	19.2	
Change in external debt	-24.6			-19.3	-7.8	-2.0	-2.6	-2.0	-2.5		-1.0	-0.4	
Identified net debt-creating flows	-14.6			5.9	5.3	4.2	5.6	6.0	5.8		6.4	6.2	
Non-interest current account deficit	**9.5**	**11.6**	**3.5**	**11.4**	**10.3**	**9.0**	**9.8**	**10.0**	**9.6**		**9.5**	**8.7**	9.3
Deficit in balance of goods and services	2.5			2.8	2.6	2.6	4.5	5.6	5.8		6.7	6.2	
Exports	37.6			34.6	31.4	31.2	28.6	27.0	26.2		24.1	22.4	
Imports	40.0			37.5	34.0	33.8	33.1	32.6	31.9		30.8	28.6	
Net current transfers (negative = inflow)	-4.1	-5.3	1.2	-4.1	-3.5	-3.3	-3.3	-3.3	-3.3		-3.2	-3.0	-3.1
Other current account flows (negative = net inflow)	11.1			12.6	11.2	9.7	8.6	7.7	7.1		6.0	5.4	
Net FDI (negative = inflow)	**-4.4**	**-4.2**	**1.5**	**-3.8**	**-2.9**	**-2.7**	**-2.1**	**-2.1**	**-2.0**		**-1.7**	**-1.4**	-1.6
Endogenous debt dynamics 2/	**-19.7**			**-1.7**	**-2.1**	**-2.1**	**-2.0**	**-1.9**	**-1.8**		**-1.4**	**-1.1**	
Contribution from nominal interest rate	1.2			0.7	0.4	0.3	0.3	0.3	0.2		0.1	0.0	
Contribution from real GDP growth	-4.0			-2.4	-2.5	-2.4	-2.3	-2.1	-2.0		-1.5	-1.1	
Contribution from price and exchange rate changes	-17.0			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-9.9**			**-25.2**	**-13.2**	**-6.2**	**-8.2**	**-8.1**	**-8.3**		**-7.4**	**-6.7**	
o/w exceptional financing	-4.9			-7.0	-2.4	-2.2	-1.9	-1.7	-1.6		-1.1	0.0	
NPV of external debt 4/	28.2			22.8	19.8	18.9	17.5	16.4	15.5		13.2	10.2	
In percent of exports	75.1			65.8	63.0	60.6	61.2	60.9	59.2		54.7	45.4	
NPV of PPG external debt	**28.2**			**22.8**	**19.8**	**18.9**	**17.5**	**16.4**	**15.5**		**13.2**	**10.2**	
In percent of exports	75.1			65.8	63.0	60.6	61.2	60.9	59.2		54.7	45.4	
Debt service-to-exports ratio (in percent)	23.0			5.0	3.3	3.8	5.3	4.8	4.1		1.2	1.4	
PPG debt service-to-exports ratio (in percent)	23.0			5.0	3.3	3.8	5.3	4.8	4.1		1.2	1.4	
Total gross financing need (millions of U.S. dollars)	746.3			641.1	700.2	670.4	885.9	960.8	977.9		1389.7	3047.1	
Non-interest current account deficit that stabilizes debt ratio	34.0			30.7	18.1	10.9	12.4	12.0	12.1		10.6	9.1	
Key macroeconomic assumptions													
Real GDP growth (in percent)	5.4	3.1	3.5	4.3	6.0	6.0	6.0	6.0	6.0	5.7	6.0	6.0	6.0
GDP deflator in US dollar terms (change in percent)	21.8	2.6	12.7	21.0	14.7	1.5	2.2	1.4	2.1	7.2	2.6	2.6	2.6
Effective interest rate (percent) 5/	1.7	3.0	0.9	1.2	1.0	0.8	0.8	0.7	0.6	0.9	0.2	0.0	0.2
Growth of exports of G&S (US dollar terms, in percent)	63.7	8.0	24.4	16.3	10.2	7.0	-0.7	1.5	4.9	6.5	8.0	8.0	7.7
Growth of imports of G&S (US dollar terms, in percent)	21.1	6.2	14.9	18.0	10.2	7.1	6.0	6.0	6.0	8.9	8.0	8.0	8.0
Grant element of new public sector borrowing (in percent)	…	…	…	55.2	49.1	53.9	56.9	56.9	56.9	54.8	56.9	56.9	56.9
Memorandum item:													
Nominal GDP (millions of US dollars)	5429.3			6850.4	8331.5	8960.5	9710.9	10434.9	11295.7		17218.8	40011.6	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are derived over 1994-2004.

Table A2. Zambia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	**2015**	2025
NPV of debt-to-GDP ratio								
Baseline	23	20	19	17	16	16	**13**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	23	23	23	21	20	19	**17**	15
A.2 Additional borrowing for power sector investment 2/	23	20	19	21	22	24	**19**	10
A3. MDRI Scenario	23	4	5	5	5	5	**7**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	23	21	21	20	19	18	**15**	12
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	23	23	28	26	25	23	**19**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	23	25	27	25	24	22	**19**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	23	20	18	17	16	15	**13**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	23	27	33	30	29	27	**23**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	23	27	26	24	22	21	**18**	14
NPV of debt-to-exports ratio								
Baseline	66	63	61	61	61	59	**55**	45
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	66	72	72	74	74	74	**72**	65
A.2 Additional borrowing for power sector investment 2/	66	63	61	72	83	90	**79**	46
A3. MDRI scenario	66	14	15	16	18	20	**27**	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	66	63	61	61	61	59	**55**	45
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	66	97	151	154	155	152	**137**	94
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	66	63	61	61	61	59	**55**	45
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	66	62	59	59	59	57	**53**	45
B5. Combination of B1-B4 using one-half standard deviation shocks	66	79	97	99	99	97	**88**	65
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	66	63	61	61	61	59	**55**	45
Debt service ratio								
Baseline	5	3	4	5	5	4	**1**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	5	4	5	7	6	5	**3**	4
A.2 Additional borrowing for power sector investment 2/	5	3	4	5	6	5	**5**	2
A3. MDRI Scenario	5	3	2	2	1	1	**1**	1
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	5	3	4	5	5	4	**2**	3
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	5	4	7	10	9	8	**5**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	5	3	4	5	5	4	**2**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	5	3	4	5	5	4	**2**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	5	4	5	7	7	6	**3**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	5	3	4	5	5	4	**2**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	56	56	56	56	56	56	**56**	56

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes borrowing of $1 billion on commercial terms over 2008-11 for investment in electrecity generation.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A3. Zambia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002-2025
(In percent of GDP, unless otherwise indicated)

	Actual			Estimate	Projections							
	2002	2003	2004	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025
Public sector debt 1/	128.9	126.8	92.6	68.5	60.9	57.0	53.2	50.5	47.1		36.0	27.8
o/w foreign-currency denominated	104.3	99.7	70.3	49.4	44.5	42.0	39.5	37.4	34.8		27.0	21.8
Change in public sector debt	-19.9	-2.2	-34.1	-24.2	-7.6	-3.9	-3.8	-2.8	-3.4		-1.4	-0.5
Identified debt-creating flows	-11.0	-13.0	-20.6	-18.1	-4.9	-2.4	-2.1	-1.4	-1.6		-0.9	-0.4
Primary deficit	1.0	2.2	-0.6	-0.1	0.8	0.9	0.6	0.8	0.9	0.7	0.2	0.5
Revenue and grants	26.2	24.9	23.8	23.9	23.6	23.7	23.9	24.0	24.0		25.0	25.0
of which : grants	8.3	7.0	5.5	6.0	6.2	6.2	6.0	6.1	6.1		5.0	5.0
Primary (noninterest) expenditure	27.2	27.1	23.2	23.8	24.4	24.6	24.5	24.8	24.9		25.2	25.5
Automatic debt dynamics	-9.0	-14.7	-20.1	-17.3	-4.9	-2.5	-2.2	-1.6	-1.9		-1.7	-1.4
Contribution from interest rate/growth differential	-4.7	-6.3	-7.3	-4.5	-1.9	-1.8	-1.8	-1.5	-1.5		-1.4	-1.2
of which : contribution from average real interest rate	0.1	0.0	-0.8	-0.7	2.0	1.7	1.4	1.5	1.3		0.7	0.4
of which : contribution from real GDP growth	-4.8	-6.3	-6.5	-3.8	-3.9	-3.4	-3.2	-3.0	-2.9		-2.1	-1.6
Contribution from real exchange rate depreciation	-4.3	-8.4	-12.7	-12.8	-3.0	-0.7	-0.4	-0.1	-0.4		...	...
Other identified debt-creating flows	-3.0	-0.5	0.1	-0.8	-0.8	-0.8	-0.5	-0.5	-0.6		0.5	0.5
Recognition of implicit or contingent liabilities (Pensions)	0.0	0.0	0.0	0.5	0.5	0.5	0.5	0.5	0.5		0.5	0.5
Debt relief (HIPC and other)	-3.0	-1.5	-0.3	-1.6	-1.5	-1.5	-1.2	-1.3	-1.3		0.0	0.0
Other (financial restructuring)	0.0	1.0	0.4	0.3	0.2	0.2	0.2	0.2	0.2		0.0	0.0
Residual, including asset changes	-8.9	10.9	-13.6	-6.1	-2.7	-1.4	-1.6	-1.4	-1.8		-0.5	-0.1
NPV of public sector debt	...	71.5	50.6	41.1	36.8	34.3	31.7	29.9	28.1		23.0	17.5
o/w foreign-currency denominated	55.1	44.4	28.2	22.1	20.4	19.3	17.9	16.9	15.9		14.0	11.5
o/w external	55.1	44.4	28.2	22.1	20.4	19.3	17.9	16.9	15.9		14.0	11.5
Gross financing need 2/	6.3	6.6	15.5	9.1	8.0	7.1	6.7	6.3	6.0		3.8	3.0
NPV of public sector debt-to-revenue ratio (in percent) 3/	...	287.0	212.4	172.3	156.0	144.5	132.7	124.5	117.2		91.9	70.2
o/w external	...	178.2	118.5	92.6	86.6	81.4	75.0	70.2	66.1		55.9	46.2
Debt service-to-revenue ratio (in percent) 3/ 4/	15.5	15.5	45.7	15.9	12.7	10.3	10.9	9.9	8.9		4.1	3.3
Primary deficit that stabilizes the debt-to-GDP ratio	20.9	4.3	33.5	24.1	8.4	4.7	4.4	3.6	4.3		1.6	1.0
Key macroeconomic and fiscal assumptions												
Real GDP growth (in percent)	3.3	5.1	5.4	4.3	6.0	6.0	6.0	6.0	6.0	5.7	6.0	6.0
Average real interest rate on domestic currency debt (in percent)	0.4	0.0	-3.2	-3.1	3.0	5.7	5.7	6.5	5.9	4.0	3.9	3.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-3.6	-8.4	-13.5	-19.0	-6.2	-1.5	-1.0	-0.3	-1.1	-4.8	-0.9	-0.9
Inflation rate (GDP deflator, in percent)	19.9	19.8	20.1	18.9	11.5	4.6	4.1	3.6	4.4	7.9	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-4.9	4.6	-9.5	6.8	8.8	7.0	5.4	7.4	6.6	7.0	6.2	6.1

Sources: Zambia authorities; and Fund staff estimates and projections.

1/ Gross debt of Central Government Overall Operations, including IMF claims to the Bank of Zambia. It includes the audited stock of statutory pension arrears, which was K 322 bn by end-March 2005, and the end-year stock of domestic arrears.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Zambia: Sensitivity Analysis for Key Indicators of Public Debt 2005-2025

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Baseline	41	37	34	32	30	28	23	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	40	35	34	32	31	30	35	45
A2. Primary balance is unchanged from 2004	42	34	29	24	21	17	9	-2
A3. Permanently lower GDP growth 1/	42	36	33	30	27	25	22	26
A4. MDRI Scenario	41	21	20	18	18	18	16	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	42	37	36	34	32	31	31	35
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	42	38	38	34	32	29	23	18
B3. Combination of B1-B2 using one half standard deviation shocks	42	38	37	34	31	28	21	14
B4. One-time 30 percent real depreciation in 2006	42	44	40	36	33	30	23	16
B5. 10 percent of GDP increase in other debt-creating flows in 2006	42	46	42	38	36	33	27	21
NPV of Debt-to-Revenue Ratio 2/								
Baseline	172	156	144	133	124	117	92	70
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	167	150	140	132	127	122	134	166
A2. Primary balance is unchanged from 2004	174	144	122	103	87	72	37	-8
A3. Permanently lower GDP growth 1/	176	152	137	123	114	106	89	103
A4. MDRI Scenario	172	87	80	75	73	72	64	60
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	176	158	149	138	132	126	120	139
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	176	163	158	143	131	122	93	73
B3. Combination of B1-B2 using one half standard deviation shocks	176	162	156	139	126	115	82	54
B4. One-time 30 percent real depreciation in 2006	176	188	169	150	137	126	92	62
B5. 10 percent of GDP increase in other debt-creating flows in 2006	176	193	176	160	148	138	107	82
Debt Service-to-Revenue Ratio 2/								
Baseline	16	13	10	11	10	9	4	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	16	13	13	16	17	19	25	39
A2. Primary balance is unchanged from 2004	16	12	8	8	6	4	-5	-16
A3. Permanently lower GDP growth 1/	16	13	11	13	13	13	11	20
A4. MDRI Scenario	16	12	8	7	6	5	3	2
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	16	13	13	16	17	18	21	33
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	16	13	15	20	20	20	14	12
B3. Combination of B1-B2 using one half standard deviation shocks	16	13	14	18	17	17	10	5
B4. One-time 30 percent real depreciation in 2006	16	13	11	13	12	12	8	7
B5. 10 percent of GDP increase in other debt-creating flows in 2006	16	13	26	27	26	26	19	15

Sources: Zambia authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

2/ Revenues are defined inclusive of grants.